Nostrian Conquest

Player Manual

A from-scratch Rust recreation inspired by the classic 1990s BBS door game Esterian Conquest.

Built on Nostr for decentralized play. All code, UI, and assets are original.

Not affiliated with any original release. Created for fun and retro preservation.

Revision date: April 11, 2026

Version 1.0.0-beta.2 — Beta

Contents and Appendices

1. Introduction

Beyond the mapped frontiers of the old Nostrian dominion lies a small galaxy of contested solar systems. The old masters are gone. Their stations are silent, their patrols vanished, and their subjects left with fleets, industry, and enough knowledge to build empires.

You rise as one of the new Star Masters. From a single world and a few small fleets, you must tax, build, scout, bargain, threaten, and strike before rival powers can do the same. Some systems will join your banner willingly. Others will require persuasion from orbit.

Each maintenance marks the passage of a year. In that span, fleets cross the dark between stars, colonies grow or starve, alliances turn cold, and wars are decided by distance, industry, mathematics, and will.

Nostrian Conquest is a from-scratch Rust recreation inspired by the classic 1990s BBS door game Esterian Conquest. All code, UI, and assets in this edition are original, and it is not affiliated with any original release.

In homage to the 1990s door-game pioneers and to the ancient dreamers, strategists, and storytellers whose visions of galactic dominion still light the way among these stars.

2. Connecting to a Game

2.1. Three Ways to Play

Nostrian Conquest supports three practical ways to play.

The normal public path is **Nostr**. In that mode you run `nc-connect`, your sysop gives you an invite code, and `nc-connect` opens the live `nc-game` session for you. This is the recommended path for public multiplayer.

The direct private path is **Localhost**. In that mode you or your sysop runs `nc-game` directly on the same machine. Use this for solo play, hotseat testing, and trusted same-machine sessions. If you are setting that up yourself, see the **Sysop Manual** section **Localhost Session Setup**.

The classic path is **BBS**. In that mode you log into a bulletin board with a terminal client and launch the game from the doors menu. The sysop stages `nc-door` behind the BBS. If you are the sysop, see the **Sysop Manual** section **BBS Door Setup**.

2.2. Finding New Games

If you do not already have an invite code, start at nostrian-conquest.com. That landing page points to the current public meeting places for live campaigns and player announcements.

Right now, the main public meeting place is the Discord channel at discord.gg/FMr8sfBa. The landing page also points to the `NC Sysop` Nostr/Primal presence for direct contact.

Once a sysop has a seat ready for you, he gives you an invite code in the normal `amber-river@relay.example.com` form.

2.3. Joining a Game

Your sysop will give you an invite code like `amber-river@relay.example.com`. Here is the normal picker flow:

1. Run `nc-connect`.
2. Press `N` to join a new game.
3. Paste your invite code and press Enter.

In the packaged GUI, paste works with `Command-V` on macOS, `Ctrl-V`, `Ctrl-Shift-V`, `Shift-Insert`, or right-click.

That is all. `nc-connect` handles your identity and opens your `nc-game` session. The invite already carries the relay host, and `nc-connect` discovers the rest from there. Your seat is not claimed until you actually save your empire name in the game. As soon as you name your empire and claim the invite, `nc-connect` downloads the campaign starmap bundle so maps are available locally for turn 1. The downloaded starmap bundle and CSV sheets are saved in your Documents `nc/maps` folder by default. Later, press `M` in the picker to change the default maps folder or re-download the bundle for the currently selected game.

One NC keychain identity can hold only one seat in a hosted game. If you have already completed the first join for that game and later delete the local picker row, press `N`, paste the same invite again, and reconnect with that same identity. The original invite still belongs only to that identity; a different identity cannot use it to take over the seat.

2.4. nc-connect Setup

Get the current `nc-connect` build from the repo's GitHub Releases page. Public player packages are available for Windows x64, Linux x64, and macOS Apple Silicon. Keep this manual with it. This package is for the Nostr path. Localhost and BBS play do not use `nc-connect` as the game client.

The public Nostrian player package contains only Nostrian binaries, manuals, and support files. It does not bundle preserved Esterian Conquest executables or manuals.

2.4.1. Windows

If your sysop gives you the Windows `.zip` build, extract it to a folder of your choice. Double-click `nc-connect.exe` to launch the normal player window. No installation required.

On first use, `nc-connect` creates an encrypted keychain. You choose one keychain password for the machine. That password protects your local identities. If you lose it, the client cannot recover your keychain for you.

Your sysop gives you one invite code in the form `amber-river@relay.example.com`. Paste it when prompted.

2.5. Keychain Management

The packaged GUI keeps one active identity at a time.

From the main `nc-connect` picker:

- `Y` opens the keychain screen
- `I` shows identity info

From the keychain screen:

- `R` replaces the current identity: paste an existing `nsec`, or leave the field blank to generate a fresh one
- `Enter` shows the full `npub` and `nsec`

The keychain is local client state. It is not the server's player list. Replacing your current identity does not move any already-claimed hosted seat. Those stay bound to the original identity until your sysop reissues the seat with a new invite.

2.5.1. Clearing Local Keychain Data

To clear your local keychain and picker cache, close `nc-connect` and delete the files manually:

- **Windows:** `%LOCALAPPDATA%\nc\` (e.g. `C:\Users\<you>\AppData\Local\nc\`)
- **macOS:** `~/Library/Application Support/nc/`
- **Linux:** `~/.local/share/nc/`

Delete `keychain.kdl` to remove your identity, or `cache.kdl` to clear the picker game list. Removing `keychain.kdl` is permanent; a fresh identity will be created on next launch.

2.6. Relay Configuration

Press lowercase `r` in the main `nc-connect` menu to open the relay manager. That screen lets you add relays, edit saved relay entries, mark a default relay, and inspect which joined games use each relay.

Press uppercase `R` in the main game list to edit the relay for the currently selected joined game only. That is mainly useful when fixing an older cached entry or moving one specific game to a different relay.

After a successful join, `nc-connect` caches the exact relay for that game, so most reconnects do not need relay entry again.

2.7. Refreshing Game Info

Press `Space` on a selected joined game to refresh its cached hosted metadata. This asks the hosted service for the latest game name, seat, player-name, and relay details for that one entry without opening a play session.

3. Quick Start: Gameplay Basics

3.0.1. Your Objective

Your objective is simple: become Emperor by dominating rivals or eliminating every serious threat.

You begin with one planet at 100 production and four fleets — two carrying an ETAC and cruiser for colonization, and two single-destroyer scouts. Set a tax rate around 50–65% on your homeworld, and use the revenue to build ships, armies, batteries, and starbases. Keep taxes low on new colonies so they develop quickly.

Each round represents one year. You submit orders during the year, and maintenance resolves all empires simultaneously on an internal 52-week timeline. Reports are dated as `Stardate WK/YYYY` (week/year).

3.0.2. Turn Progression

There is no "End Turn" button in Nostrian Conquest. Because all players submit orders simultaneously, the game does not wait for a player to explicitly pass the turn. Instead, you simply set your economy, assign fleet missions, and log off. The sysop (or an automated server schedule) determines when the "maintenance cycle" runs. When it does, the game engine processes everyone's orders, resolves battles, advances time by one year, and generates a new set of reports for you to read the next time you connect.

3.0.3. Assets You Can Build

The table below lists everything your planets can produce. **AS** is Attack Strength (firepower) and **DS** is Defense Strength (hull/shields).

Item	**Cost**	**Size**	**Speed**	**AS**	**DS**	**Purpose**
Destroyer	05	S	6	10	05	Combat, scouting, defense
Cruiser	15	M	5	30	30	Balanced combat
Battleship	45	L	4	90	100	Heavy combat
Scout	15	S	6	00	10	Reconnaissance
Troop Transport	05	M	5	00	10	Deliver armies
ETAC	20	L	3	00	20	Colonize raw planets (reusable)
Ground Battery	20	L	–	90	20	Planetary defense
Army	02	S	–	10	10	Surface defense and capture
Starbase	50	L	1	100	120	Defense, production boost

3.0.4. Fleet Missions (Summary)

Fleet missions fall into three categories. *One-shot* missions cause the fleet to travel, perform an action, and then revert to Hold Position — you must issue new orders afterward. *Persistent* missions remain active until you replace them or game rules invalidate them. *Hostile* missions send the fleet to a target world where it waits; the assault executes on the *next* maintenance tick after arrival, not immediately.

No	Mission	Type	Requirements	On Arrival / Completion
00	Hold position	Persistent	Any ships	Idle at current position
01	Move to sector	One-shot	Any	Reverts to Hold on arrival
02	Seek Home	One-shot	Any	Reverts to Hold; retargets if destination captured
03	Patrol	Persistent	Any	Patrols sector continuously
04	Guard starbase	Persistent	Combat ships	Escorts starbase continuously
05	Guard/blockade world	Persistent	Combat ships	Blockades world continuously
06	Bombard world	Hostile	Combat ships	Bombards when in orbit at tick start
07	Invade world	Hostile	Combat + loaded transports	Invades when in orbit at tick start
08	Blitz world	Hostile	Loaded transports (combat recommended)	Blitzes when in orbit at tick start
09	View	One-shot	Any	Scans from system edge, reverts to Hold
10	Scout sector	Persistent	At least one scout	Reports each turn while on station
11	Scout system	Persistent	At least one scout	Reports each turn while on station
12	Colonize world	One-shot	At least one ETAC	Colonizes if unowned; ETAC survives for reuse
13	Join Fleet	Persistent	Any	Chases host until merge; abandons if host lost
14	Rendezvous	Persistent	Any	Waits at sector; lowest fleet ID becomes host
15	Salvage	One-shot	Any	Scraps ships for ~50% value, reverts to Hold

4. Forces at Your Command

You control five major force types: planets, ships, starbases, armies, and ground batteries.

4.0.1. Planets

Planets are the foundation of your empire. Each one has a **Potential Production** — the ceiling it can reach at full efficiency, ranging from 10 to 150 across the galaxy — and a **Production**, which is what the planet can actually deliver right now. Production grows toward Potential over time, and the speed of that growth depends heavily on your tax rate.

Taxes convert Production into spendable build points each year. See Section 5 for the full tax rate, growth, and starbase economics model.

4.0.2. Ships

Ships operate in fleets. A fleet always moves at the speed of its slowest member and can hold anywhere from one to **3,000 ships** of mixed types.

Ship Type	Cost	Speed	Size	AS	DS	Tactical Role
Destroyer	05	6	S	10	05	Fast, cheap screen. Escapes heavy fleets. Good for scouting.
Cruiser	15	5	M	30	30	Balanced fighter. About 3x the power of a destroyer.
Battleship	45	4	L	90	100	Heavy firepower anchor. About 3x the power of a cruiser. Slow but durable.
Scout	15	6	S	00	10	Stealthy spy. A lone scout is hardest to detect.
Transport	05	5	M	00	10	Unarmed. Carries one army. Essential for conquest.
ETAC	20	3	L	00	20	Colony ship. Reusable — survives colonization.

4.0.3. Starbases

Starbases are large space fortresses (Cost: 50, AS: 100, DS: 120) that serve dual roles. In orbit around a planet, they provide a defensive combat bonus and significant economic benefits — they help underdeveloped colonies grow faster at low and moderate tax rates, and they let a planet spend up to **5x** its current production on a single build when points have been accumulated (see Section 5 for details). They are not a free pass to run punitive tax rates forever. In deep space, they function as surveillance platforms with slightly more firepower than a battleship, though they move very slowly at just 1 sector per year.

Unlike ships, starbases are not assigned to fleets. They are commissioned individually from stardock and moved independently through the **Starbase Command** submenu. Older documentation sometimes says a starbase is “hauled,” but in practice this is simply the normal move order for a very slow unit, likely implying tug support rather than a separate modeled mechanic. You can order combat fleets to escort a starbase using Mission 4 (Guard Starbase), but the starbase itself remains a separate unit.

> **NOTE:** Starbases must be commissioned from stardock before they can be moved or provide orbital benefits. An uncommissioned starbase sitting in stardock has no effect.

4.0.4. Ground Forces

Armies (Cost: 2, AS: 10, DS: 10) defend your planets from invasion and are the only way to capture enemy worlds. Each troop transport carries one army, and a successful invasion requires landing enough armies to overwhelm the defending garrison.

Ground Batteries (Cost: 20, AS: 90, DS: 20) are immobile land-based cannons that offer massive firepower per cost — roughly equal to a battleship at less than half the price. During an invasion, all batteries must be destroyed before transports can land. During a blitz, surviving batteries fire directly on descending transports, inflicting heavy losses. If a planet is captured via blitz, any surviving batteries transfer intact to the new owner.

5. Economy and Taxes

Your empire runs on production. Every owned planet generates tax revenue each maintenance cycle, and that revenue pays for ships, defenses, and expansion. Managing the tension between short-term revenue and long-term growth is one of the deepest strategic challenges in the game.

5.0.1. Key Terms

Every planet has a **Potential Production** — the maximum productive capacity it can ever reach — and a **Production**, which is what it can deliver right now. Production grows toward Potential over time. Your empire's **Empire Revenue** is the sum of tax revenue across all your planets. Any revenue a planet does not spend accumulates on it as its **Treasury** — the reserve of saved production points available to fund future builds. Each turn, how much of that treasury a planet can actually spend is limited by its **build capacity**; that per-turn spending limit is the planet's **Budget**.

5.0.2. Tax Revenue

The tax rate is empire-wide: you set one rate for all your planets. Revenue per planet per year is a fixed percentage of Production, and your empire's Empire Revenue is the sum of that revenue across all owned planets. See Section 11 for the exact formula.

5.0.3. Growth Toward Potential

Each maintenance turn, every owned planet grows its Production toward its Potential. Lower taxes produce faster growth — a planet at 30% tax develops far faster than one at 60%. Growth also slows naturally as a planet approaches its ceiling. Even at punishingly high tax rates, a planet below its Potential always grows by at least 1 point per year.

The engine computes this from the remaining gap to Potential and the tax headroom, then clamps the result so a planet below Potential always grows by at least `1` and never grows by more than the remaining gap. See Section 11 for the exact formula.

5.0.4. The 65% Tax Threshold

> **WARNING:** Setting taxes above **65%** can directly *reduce* Production on your planets, not just slow growth.

Below 65%, growth is always positive — lower is faster. Above 65%, a penalty kicks in that actively damages Production. A commissioned starbase does **not** remove this danger. Its value is that it helps young colonies grow faster before you reach punitive tax rates, and it still preserves the powerful build-capacity bonus described below.

The recommended early-game rate is around 50–65%. Drop taxes on new colonies to accelerate their development. See Section 11 for the exact penalty and yearly update formulas.

5.0.5. Starbase Economic Effects

A commissioned starbase in orbit provides two major benefits. First, the planet's yearly production growth gets a strong boost when taxes are modest — the full bonus applies at **50% tax or lower**, then tapers away as taxes climb toward **65%**. This means underdeveloped planets with starbases catch up significantly faster when you are managing them sensibly, but the bonus is not meant to offset punitive taxes. Second, the planet gains a **build capacity multiplier** — it can spend up to **5x** its Production on builds in a single turn, drawing from its treasury. Without a starbase, a planet can only spend up to 1x its Production per turn. See Section 11 for the exact bonus taper and build-capacity formulas.

> **NOTE:** These bonuses require an active, commissioned starbase in orbit — not an uncommissioned starbase sitting in stardock.

5.0.6. Treasury

Tax revenue that a planet does not spend accumulates in its **Treasury** — the planet's savings reserve. The treasury is what allows starbase worlds to execute large builds: up to 5x Production in a single turn. Each turn, a planet's **Budget** is the lesser of its treasury and its build capacity. That is what the build screen shows as **BUDGET** — how many production points remain uncommitted this turn. When maintenance processes a build queue, only the points actually spent that year are deducted from the treasury; unfinished builds keep their remaining cost for later turns.

5.0.7. Newly Colonized Planets

A freshly colonized planet starts with Production far below its Potential and with no treasury. It does not collect revenue or growth on the same maintenance turn that establishes the colony. Growth begins on later turns, and because tax revenue is credited before growth is applied, a new colony can remain at zero budget for multiple turns even at reasonable tax rates. Keep taxes low on new colonies so they develop quickly.

5.0.8. Conquered Planets

When you capture an enemy planet by invasion or blitz, the planet's industry needs approximately **two turns** before it is fully converted and begins producing tax revenue for your empire. Plan your logistics around this delay.

6. Combat Mechanics

While battle reports are simple summaries, the engine uses a sophisticated system behind the curtain. Inspired by the simultaneous-resolution combat model in Mark Herman's tabletop wargame *Empire of the Sun*, the Rust engine is deterministic and simultaneous rather than relying on opaque random number generation or arbitrary ship-vs-ship duels.

6.0.1. The Rules of Battle

There is no "first strike" advantage. Each round, the total **Attack Strength (AS)** of each fleet is calculated and inflicted upon the enemy at the same instant. Rounds repeat until one side is destroyed, disengages, or only one hostile force remains.

Damage reduces ships from "Nominal" to "Crippled" status before destroying them. All nominal ships must be crippled before any crippled ship is destroyed, and surviving crippled ships are repaired automatically after battle. Hits always target the combat line first — destroyers, cruisers, battleships, and starbases. Scouts, transports, and ETACs are protected as long as any combat-line ships remain, which means your non-combat vessels survive as long as your warships hold the line.

Mixed fleets containing destroyers, cruisers, *and* battleships receive a **combined arms bonus** that improves their tactical effectiveness compared to single-type fleets. Always mix your composition when possible. A defending starbase at its own world provides an additional combat bonus to the defender. In a draw, the defender wins. See Section 12 for the exact ROE thresholds, combat values, force-ratio columns, CRT table, and assault formulas used by the current engine.

6.0.2. Tactical Roles and Split Fire

Combat is more than just a numbers game. Different ship classes perform specific tactical roles based on how they deliver fire. Hits generated by a task force are split into two pools:

Suppression Fire (Cruisers and Battleships): **These heavy vessels provide volume and suppression. Their hits are dispersed across the enemy fleet, reducing nominal ships to a** crippled **state. This effectively knocks enemy guns offline and wins the immediate field, but it does not always result in permanent kills. Execution Fire (Destroyers and Starbases):** These units use **focus fire** to eliminate specific targets. Their hits bypass the crippled state and allocate directly to the **destroyed** pool, paying the full 2x DS cost to blow ships up one by one.

This creates a deadly synergy. A commander uses his heavy ships to suppress and soften the enemy line, while his destroyers and starbases pick off the survivors.

> **NOTE:** All planetary return fire from ground batteries is treated as **Execution Fire**. Fortified worlds do not "soften" an attacker; they destroy him.

6.0.3. Fleet Limits

A fleet can contain as few as one ship and as many as **3,000 ships** of mixed types. A fleet always moves at the speed of its slowest member.

6.0.4. Rules of Engagement (ROE)

You assign an ROE level (0–10) to control when your fleet voluntarily engages hostile forces. ROE is a deterministic commitment rule based on force ratios, not random chance.

Before any fire is exchanged, your fleet performs a **pre-combat sensor check**. If the enemy force is overwhelming and violates his ROE, the commander will abort the engagement and "seek home" immediately. This clean retreat happens before the battle begins, allowing him to scout safely without being forced into a suicidal withdrawal exchange.

Note: Sensor checks do not trigger if you are forced into engagement (such as entering a defended system) or if your fleet is in a Guard or Incumbent role.

The full ROE threshold table is collected in Section 12.

Non-combat fleets (scouts, transports, ETACs only) are treated as ROE 0 automatically.

6.0.5. Withdrawal and Retreat

> **IMPORTANT:** Low ROE does not guarantee safety once combat begins.

If you choose to engage (or are forced to), your fleet is committed to the battle for a minimum of **three rounds**. ROE-based bailing is disabled until Round 4. This ensures that fleets trade meaningful blows before one side loses his nerve.

A fleet that breaks off after Round 3 does not escape cleanly. It suffers a **withdrawal exchange** — the enemy fires on your retreating fleet, and your fleet fires back at reduced effectiveness. Only after absorbing that exchange does the fleet actually retreat and abort its current mission. After each round of combat from Round 4 onward, surviving fleets re-check their ROE. If the post-loss ratio no longer meets his threshold, the commander attempts to disengage and suffers the withdrawal exchange.

6.0.6. Planetary Combat

When fleets attack planets through bombardment, invasion, or blitz, different rules apply. Ground batteries are the planet's shield wall — while they stand, they draw orbital fire and shoot back, protecting armies, production, and industry behind them. Only combat ships — destroyers, cruisers, and battleships — contribute bombardment firepower. Scouts, transports, and ETACs do not.

> **WARNING:** Planetary defenses use **focus fire**. Hits from ground batteries bypass the crippled state and directly destroy ships, making them far more lethal than dispersed ship-to-ship fire. A commander who brings a large fleet to orbit a fortified world will find his ships being picked off one by one by concentrated ground fire.

Each bombardment turn resolves three rounds of fire. In rounds 1 and 2, your ships trade fire with the planet's batteries. Hits land on stardock contents first, then batteries, but armies, stored goods, and factories are shielded. In round 3, if all batteries have been destroyed, your remaining firepower breaks through to armies, stored goods, and factories. If batteries still stand, round 3 is another suppression exchange and the planet's vulnerable assets survive another turn.

This means a well-defended planet takes sustained bombardment over multiple turns before you touch its production. Build batteries to buy time; bring heavy fleets to break through faster.

See Section 7 for the detailed mechanics of bombardment, invasion, and blitz missions. The exact bombardment weights, batteries-only return-fire rule, and combat tables are in Section 12.

7. Missions and Orders

A fleet always has exactly one standing order. If you issue a new order before maintenance, it replaces whatever the fleet was doing. Missions fall into three categories. *One-shot* missions cause the fleet to travel, perform an action, and revert to Hold Position — you must issue new orders afterward. *Persistent* missions remain active until you replace them or game rules invalidate them (for example, a Join mission is abandoned if the host fleet is destroyed). *Hostile* missions send the fleet to a target world where it waits; the assault executes on the *next* maintenance tick after arrival, not immediately. Plan accordingly.

No	Mission	Type	Description
00	None	Persistent	Hold position at current location.
01	Move Fleet	One-shot	Travel to a specified sector, then revert to Hold.
02	Seek Home	One-shot	Return to nearest owned planet; retargets if that planet is captured en route.
03	Patrol	Persistent	Move within sector deep space to intercept enemies.
04	Guard Starbase	Persistent	Escort a starbase; fight alongside it.
05	Blockade	Persistent	Prevent access to a planet. Stops enemy launches and landings.
06	Bombard	Hostile	Damage factories, batteries, armies, stardock contents, and production.
07	Invade	Hostile	Suppress batteries, bombard, then land armies. Requires loaded transports.
08	Blitz	Hostile	Drop armies immediately, dodging batteries. High army risk.
09	View	One-shot	Long-range scan of owner and production from system edge.
10	Scout Sector	Persistent	Passive stealth surveillance of sector deep space (requires Scout).
11	Scout System	Persistent	Active spy mission into a solar system (requires Scout).
12	Colonize	One-shot	Terraform and claim unowned planet (requires ETAC). ETAC survives for reuse.
13	Join Fleet	Persistent	Chase and merge with target fleet. Abandons if host is destroyed.
14	Rendezvous	Persistent	Meet other fleets at sector. Lowest fleet ID becomes host.
15	Salvage	One-shot	Scrap ships at planet for ~50% of build cost.

7.0.1. Mission Details

7.0.1.1. One-Shot Missions

Mission 1: Move to Sector. A simple transit order. The fleet travels to the destination sector at the speed of its slowest ship, then stops and reverts to Hold Position. You must issue new orders if you want it to do anything else.

Mission 2: Seek Home. The fleet travels to the nearest planet you own. If that planet is captured while the fleet is en route, it automatically redirects to the next nearest friendly planet. On arrival, it reverts to Hold Position.

Mission 9: View a World. A safe, long-range scan. The fleet approaches the edge of the target system, scans for owner and production data, and immediately backs off into deep space. It reverts to Hold Position after reporting.

Mission 12: Colonize a World. Requires at least one ETAC (Environmental Transformation And Colonization ship). If the planet is unowned, it is terraformed and claimed. The new colony starts with one garrison army, no treasury, and very low current production. It does not receive revenue or growth until later maintenance turns, and then develops faster under low taxes. The ETAC is not consumed — it survives and can colonize additional planets. If the planet is already owned, the ETAC aborts, reports the owner's identity and production potential, writes that partial result into your Total Planet Database, and waits for new orders.

Mission 15: Salvage. The fleet travels to the specified planet and scraps its ships for approximately **50%** of the original build cost, returned to that planet's treasury. It reverts to Hold Position after scrapping.

7.0.1.2. Persistent Standing Missions

Mission 0: Hold Position. The default idle state. The fleet stays at its current location and takes defensive action based on ROE if hostile fleets approach.

Mission 3: Patrol a Sector. The fleet moves within deep space to intercept enemies passing through. If the patrol spots anything, it sends a report, and it engages based on your ROE settings. The mission remains active until you assign a new one.

Mission 4: Guard Starbase. The fleet escorts a starbase and fights alongside it. Be aware that if your fleet has a high ROE, it may break formation to chase enemy fleets entering the system, leaving the starbase temporarily vulnerable.

Mission 5: Guard/Blockade a World. A strategy of denial. The blockade stops enemy fleets from using the planet, intercepts ships launching from stardock, and paralyzes the enemy's ability to deploy forces from that world. It remains active until you assign a new mission.

Missions 10 and 11: Scout Sector / Scout System. Both require at least one Scout ship, and a fleet consisting of a single Scout is the least likely to be detected. Scout Sector is a passive, stealthy patrol — unlike Mission 3, the fleet will not engage enemies but instead relies on stealth to observe traffic without being seen. Scout System is an active spy run where the Scout penetrates the system to report on ground batteries, armies, current production, stardock contents, and orbiting fleets. Both are persistent missions: the fleet remains on station and generates a new report each maintenance turn for as long as it stays at its assigned sector. If you identify an **In Civil Disorder** fleet in a system that still belongs to that disorder state, the Total Planet Database marks that world's owner as `ICD` even before a full scout/view report is earned.

Missions 13 and 14: Fleet Coordination. Mission 13 (Join Fleet) causes the fleet to chase a specific host fleet and merge with it when they meet. If the host is destroyed before they rendezvous, the joining fleet abandons the mission. Mission 14 (Rendezvous) sends multiple fleets to a sector where the fleet with the lowest Fleet ID becomes the host of the combined force. The rendezvous point remains active so additional fleets can keep merging there.

7.0.1.3. Hostile (Delayed-Resolution) Missions

> **IMPORTANT:** Hostile missions require the fleet to be **in orbit at the start of maintenance** to execute. A fleet that arrives at the target world this turn will carry out its assault **next turn**. This one-turn delay is a critical tactical consideration — defend accordingly.

Mission 6: Bombard a World. Only destroyers, cruisers, and battleships contribute bombardment firepower — scouts, transports, and ETACs do not. Each turn your fleet bombards, the engine runs three rounds of fire. In rounds 1 and 2, your ships exchange fire with ground batteries — hits destroy stardock contents first, then batteries, but armies, stored goods, and industry are shielded behind the battery wall. In round 3, if batteries have been eliminated, your firepower breaks through to armies, stored goods, and finally industry. If batteries still stand after round 2, round 3 is another suppression exchange and the planet's production survives. Bombardment persists each turn until you issue new orders. Use it to grind down a world's defenses before invasion, or to deny resources to an enemy over time.

Mission 7: Invade a World. A three-stage deliberate assault. First, combat ships exchange fire with ground batteries in orbital suppression — transports cannot land until all batteries are destroyed. Once batteries are gone, surviving combat ships fire on the defending armies to soften resistance before landing. Unlike bombardment, invasion softening targets armies only — industry and stored goods are preserved because the goal is to capture the planet with its production intact. However, orbital softening can destroy at most **half** of the defending army stack; at least half of the original defenders must still be fought on the ground. Finally, transports land their armies and the ground battle continues in repeated simultaneous rounds until one side is wiped out. The defender fights from cover with a defensive edge, and ground combat cannot end in a draw — if a final exchange would wipe out both sides, the larger force entering that exchange is treated as the winner, with exact equality favoring the defender. Capture requires destroying all defending armies, after which your surviving armies become the new garrison. Conquered planets need approximately two turns before they are fully converted to your production.

Mission 8: Blitz a World. Transports drop armies immediately in a fast assault that bypasses the full orbital suppression sequence. Escorting combat ships provide brief cover fire, but surviving batteries fire directly on descending transports, causing heavy losses. Once on the surface, armies fight in repeated simultaneous rounds until one side is wiped out, and the defender receives a defensive bonus throughout the ground battle. If you take the planet, surviving ground batteries transfer intact to your control. The blitz preserves industry but carries high risk to your armies and transports — a 2:1 army advantage or better is recommended. Choose blitz when the planet has few or no batteries and you want to preserve its industry, or when speed matters more than casualties.

8. Interface and Commands

The game is organized around four primary menus. From the **Main Menu**, you access General Command (**G**) for autopilot, diplomacy, and reports; Planet Command (**P**) for economy and production; Fleet Command (**F**) for ship movement and missions; Information Database (**I**) to review known planet data; and View Starmap (**V**) for a graphic map of the galaxy.

8.0.1. Visual Themes

The `nc-game` client is themable. Each campaign has a sysop-chosen default theme, and local-terminal players can open **C>olor Theme** from the Main Menu or First Time Menu to choose their own session palette from the bundled theme list. The shipped bundle includes `tokyo_night`, `mag16`, and several other built-in palettes, plus a monochrome `Mono` option in the picker. You can preview and apply these without leaving the client, and your last local theme choice is remembered for your empire in that campaign.

In BBS door mode, `nc-game` instead keeps the classic **A>nsi color ON/OFF** toggle and always begins from the bundled `mag16` theme each session so classic ANSI16 terminals get a stable palette. Pressing **A** switches between that `mag16` view and a greyscale monochrome projection for the current session. Saved local theme preferences do not apply in door mode.

8.0.2. General Command

General Command handles empire-wide administration. Autopilot (**A**) lets the computer manage your defenses if you miss turns. By default, a hosted or direct Rust campaign turns autopilot on automatically after three consecutive missed turns, and the first real return that year — logging into `nc-game` or successfully submitting a turn file — turns that inactivity-triggered autopilot back off. Manual autopilot stays manual until you change it yourself. Diplomacy (**E**) lets you declare other players as Neutral or Enemy — neutral fleets will not attack unless provoked, while enemy fleets will attack on sight based on ROE. Messages (**C**) lets you send messages to other empires.

To keep messaging civil, you may send no more than three messages to any single opponent per turn. In a four-player game, for example, that is up to twelve outgoing messages in a turn. Messages and reports are never automatically purged — they accumulate in your inbox across turns until you remove them yourself. The inbox supports type and year filters to help you find older items, and pressing **D** on a selected item prompts for deletion, defaulting to yes. General Command also offers a bulk delete to clear all messages at once. New reports and messages from the most recent maintenance turn are presented one by one through the scrolling intro review when you log in, giving you another opportunity to read and delete before reaching the main menu.

The empire rankings table shows each joined empire in one of three public states: **Active**, **MIA**, or **Defeated**. **MIA** means inactivity autopilot is currently running because that player missed three consecutive turns. **Defeated** means the empire has been eliminated from active command.

8.0.3. Defeat, Recovery, and Victory

Losing your last planet does not always defeat you immediately. If you still have a recovery path, you remain **Active** and receive a **three-turn recovery window** to reclaim a world. A recovery path means at least one loaded troop transport (TT*) somewhere in your surviving fleets, or at least one ETAC while any unowned planet still exists anywhere on the map. If you retake or colonize a world before the window expires, the countdown clears and your empire continues normally.

If your last recovery force is destroyed while you are planetless, or if the three-turn recovery window expires first, your empire becomes **Defeated**. Fleet Command sends you a final defeat report, and you receive one last review-only login so you can read the closing reports and messages. After that final review, you may no longer enter the campaign.

When you eliminate another empire, Fleet Command reports that you delivered the final blow. This is separate from overall game victory. The game ends only when one empire is recognized as Emperor and no other serious contender remains. At that point every still-playable empire receives a game-over report naming the victor. The winner may continue to log in in **survey mode** to inspect the final state of the galaxy, but may not issue orders or submit turns. All other players receive one final review-only pass and are then blocked from further play.

8.0.4. Planet Command

Planet Command controls your economy and ground operations. Tax (**T**) sets the empire-wide tax rate. From the main Planet Command menu, Scorch Earth (**S**) destroys your own industry to deny it to an invader. Build (**B**) spends production points on ships, defenses, or starbases. Commission (**C**) assigns newly built ships from stardock into active fleets. Load and Unload (**L** / **U**) move armies between the planet surface and troop transports.

The **Planet List** is the fast row-centric operations screen for owned worlds. Once you open it, the highlighted planet becomes the working row for the most common actions: Build (**B**), Auto-Commission (**A**), Commission (**C**), Load / Unload armies (**L** / **U**), and Scorch Earth (**X**). On that screen, **S** is reserved for **Sort**, while **I** or **Enter** opens planet information for the selected row.

8.0.5. Fleet Command

Fleet Command controls your ships in space. Mission (**O**) assigns missions 0–15. ROE (**C**) changes a fleet's rules of engagement. Merge (**M**) combines fleets that are in the same sector. Transfer (**T**) moves individual ships between fleets.

The **Fleet List** is also the bulk-fleet work screen. It now includes a **Sel** column. Press **Space** on a fleet row to check or uncheck it. When one or more fleets are checked, **O** assigns one mission to the checked set, **C** changes their shared ROE or speed, **M** merges the checked fleets using the lowest Fleet ID as the host, and **T** opens ship transfer for a checked pair. Row-based commands such as **Review**, **ETA**, **Detach**, **Load**, and **Unload** still use the highlighted fleet.

8.0.6. Building and Commissioning

Each planet has a **10-slot build queue**. During maintenance, a planet processes as many queued build points as its current per-turn build capacity allows. Small orders may finish in one maintenance turn, while larger ones can stay queued across multiple years until the remaining cost reaches zero. As enough points are applied to complete individual units, ships and starbases move to **Stardock** — a holding area on the planet where they sit idle and vulnerable until commissioned — even if other units from the same order remain queued. Ships are commissioned into numbered fleets, while starbases are commissioned individually and managed through their own Starbase Command submenu. Armies and ground batteries, by contrast, deploy directly to the planet surface and do not pass through stardock.

A planet without a starbase can spend up to its Production in a single turn. A planet with an orbiting starbase can spend up to **5x** its Production, drawing from its treasury (see Section 5.0.6). The build screen shows **BUDGET** — your treasury capped by build capacity, minus what you have already committed this turn.

> **NOTE:** The build queue, stardock, and treasury are all linked. If stardock is full for a given unit type, that slot stops drawing points. Blocked slots do not drain your treasury — but they also do not free capacity for other work. Commission your ships promptly to keep the pipeline moving.

> **WARNING:** Troop transports are built empty. You must manually load armies onto them before sending them into battle.

> **WARNING:** Stardock contents are a prime target for enemy bombardment. Commission your ships promptly or risk losing them before they ever see combat.

> **NOTE:** The starmap can be exported as a TXT file, a CSV grid, and a CSV details sheet for offline planning. In the recommended hosted flow, `nc-connect` downloads that static bundle automatically the first time you join. Local Rust-client play can also export it directly from the in-game starmap view.

9. Strategy

9.0.1. Early Game: The Land Grab

The opening turns are a race for territory. ETACs are your most valuable early asset — grab every raw planet you can find before your rivals do. Send single destroyers ahead as scouts to locate neighbors before they locate you. Keep your tax rate at or below 65% to avoid damaging production, and drop taxes even lower on new colonies so they develop quickly. The temptation to tax at 100% for immediate cash is strong, but it cripples long-term growth.

9.0.2. Mid-Game: Consolidation

Once the easy colonies are claimed, the game shifts to fortification and intelligence. Build starbases on your best worlds to boost both defense and production capacity. Never attack a planet blindly — use Scout ships on Mission 11 to count enemy batteries and garrison strength before committing forces. This is the “get tough” phase: when raw planets are gone, the only way to grow is war.

9.0.3. Late Game: Total War

In the endgame, fleet composition and denial matter more than raw numbers. Mix destroyers, cruisers, and battleships in every fleet to trigger the combined arms bonus and distribute damage across hull types — pure battleship fleets are expensive and miss the bonus. If you cannot hold a planet, scorch it. If you cannot take a planet, blockade or bombard it into uselessness. And in a 25-player galaxy, diplomacy is not optional: you cannot fight everyone at once. Form alliances, even temporary ones, and break them only when the timing is right.

10. Historical Context

The BBS Era (1990–1992)

The original game emerged between 1990 and 1992 as a “door game” for Bulletin Board Systems. In an era before the World Wide Web, players dialed into servers over phone lines to play strategy games against dozens of strangers. Everything ran through rigid 80x25 text terminals — every menu, every star map, every battle report squeezed into that tiny viewport. It stood alongside classics like *TradeWars 2002* and *Solar Realms Elite*, but was distinguished by its depth, its hands-off design, and the sheer scale of its campaigns.

What Made It Special

Most multiplayer games of the era demanded constant attention. Esterian Conquest was different. You checked in once a day, submitted your orders, and went about your life. Overnight, the engine processed every empire simultaneously — fleets moved, economies grew, battles resolved, and alliances were tested. When you logged in the next day, a stack of reports was waiting. Campaigns ran for months, and the stories they produced — surprise invasions, desperate blockades, betrayals at the worst possible moment — were the kind that stuck with players for years.

The Rust Port (2026)

This version is a full Rust reimplementation of the original game, rebuilt from the ground up and validated against the original binaries as an acceptance oracle. The deterministic mechanics — movement, economy, build queues, cross-file linking — were recovered from the original executables and manuals, then turned into documented engine rules. Where the original behavior was hidden, stochastic, or tied to an irreproducible internal RNG (combat resolution, AI decisions), the Rust engine substitutes its own seeded, documented, and reproducible rules that preserve the structure and spirit of the originals. The result is faithful to the manuals, preserves classic compatibility at the import/export and oracle boundary, and is honest about what was recovered versus what was rebuilt. If you played the original game on a BBS in the 1990s, it should feel right. If you are discovering it for the first time, you are playing a careful reconstruction — not a guess.

The project has now reached a real beta stage. The Rust player, connection, and sysop tools cover the core campaign workflow, and the main work from here is broad playtesting, collecting feedback, and fixing the rough edges and bugs that only show up in live games while preserving the classic experience for Nostr hosts, local players, and legacy BBS sysops.

11. Appendix A: Economy Formula Reference

This appendix collects the current engine's exact economy formulas in one place.

11.0.1. Yearly Tax Revenue

```
revenue = floor(present_production * tax_rate / 100)
```

Your empire's Empire Revenue is the sum of this value across all owned planets.

11.0.2. Base Growth Toward Potential

```
gap = potential_production - present_production
tax_headroom = 100 - min(tax_rate, 95)
base_growth = ceil(gap * tax_headroom / 400)
```

Then clamp the result so a planet below Potential always grows by at least `1` and never grows by more than the remaining `gap`.

11.0.3. High-Tax Penalty Above 65%

```
if tax_rate > 65:
penalty = ceil(present_production * (tax_rate - 65) / 500)
```

Final yearly Production is:

```
present_production = min(potential_production, present_production + growth) - penalty
```

11.0.4. Starbase Growth Bonus

A commissioned starbase boosts growth at low and moderate tax rates, but the bonus tapers away completely by `65%`.

```
bonus_percent = 50 if tax_rate <= 50
bonus_percent = floor((65 - tax_rate) * 50 / 15) if 50 < tax_rate < 65
bonus_percent = 0 if tax_rate >= 65
growth = base_growth + ceil(base_growth * bonus_percent / 100)
```

11.0.5. Build Capacity, Treasury, and Budget

Per-turn build capacity is:

```
build_capacity = present_production without starbase
build_capacity = present_production * 5 with starbase
```

A planet's **treasury** accumulates unspent production points across turns. Each turn, the **budget** is capped by build capacity:

```
budget = min(treasury, build_capacity)
```

The BUDGET field on the build screen shows how many points remain after committed queue entries are subtracted. Maintenance deducts only the build points actually processed that year; any remaining build cost stays queued for later turns, and blocked builds do not consume the treasury.

12. Appendix B: Combat Tables and Formula Reference

This appendix collects the current engine's combat reference tables in one place.

12.0.1. Rules of Engagement Thresholds

ROE	Force Requirement	Behavior
00	Never engage	**Pacifist**: Flee from all hostile fleets.
01	Enemy defenseless	**Opportunist**: Engage only if the enemy has no combat capability.
02	4:1 or better	**Very Cautious**: Engage only with overwhelming advantage.
03	3:1 or better	**Cautious**: Engage only with strong advantage.
04	2:1 or better	**Favorable**: Engage with clear superiority.
05	3:2 or better	**Confident**: Engage with moderate advantage.
06	1:1 or better	**Balanced**: Engage equal or inferior forces.
07	Even if outgunned 3:2	**Bold**: Accept moderate disadvantage.
08	Even if outgunned 2:1	**Aggressive**: Accept significant disadvantage.
09	Even if outgunned 3:1	**Reckless**: Accept severe disadvantage.
10	Always	**Suicidal**: Attack regardless of the odds.

12.0.2. Unit Combat Values

The Rust engine uses a **10x internal scale** for combat values to ensure high-precision damage resolution. This ensures that crippled light ships continue to contribute fire and that attrition is granular.

Unit	AS	DS	Notes
Destroyer	10	5	Agile glass cannon
Cruiser	30	30	Balanced brawler
Battleship	90	100	Primary battle line
Scout	0	10	Non-combat hull
Transport	0	10	Non-combat hull
ETAC	0	20	Colonization hull
Starbase	100	120	Heavy orbital defender
Battery	90	20	Planetary defense
Army	10	10	Ground combatant

12.0.3. Force Ratio to CRT Column

Force Ratio	CRT Column
`< 0.5`	Disadvantaged
`0.5 .. < 1.0`	Pressed
`1.0 .. < 1.5`	Even
`1.5 .. < 3.0`	Advantaged
`>= 3.0`	Overwhelming

12.0.4. Space / Orbital CRT

d10	Disadvantaged	Pressed	Even	Advantaged	Overwhelming
0	0.00	0.25	0.50	0.75	1.00
1	0.25	0.50	0.75	1.00	1.25
2	0.25	0.50	1.00	1.25	1.50
3	0.50	0.75	1.00	1.25	1.50
4	0.50	0.75	1.00	1.50	1.75
5	0.50	1.00	1.25	1.50	1.75
6	0.75	1.00	1.25	1.50	2.00
7	0.75	1.00	1.50	1.75	2.00
8	1.00	1.25	1.50	1.75	2.00
9	1.00	1.50	1.75	2.00	2.50

12.0.5. Column Shifts and Hit Formula

- Mixed `DD/CA/BB` fleet: `+1` CRT column
- Defending starbase in orbital combat: `+1` CRT column
- Withdrawal exchange: fixed `Pressed` column
- Final columns are clamped to the table bounds

```
hits = ceil(total_AS * CRT_multiplier)
```

An unmodified `9` on the `d10` is a critical hit and forces one extra bypass loss allocation.

12.0.6. Bombardment and Planetary Fire

Only destroyers, cruisers, and battleships contribute bombardment attack strength:

- Destroyer bombardment AS: `0.5x`
- Cruiser bombardment AS: `1.0x`
- Battleship bombardment AS: `1.5x`

Planetary return fire is:

```
return_fire_AS = battery_AS
```

Planetary return fire from batteries uses the **focus fire** rule. Hits bypass the crippled state and allocate directly to the destroyed pool by paying the full `2x DS` cost per ship.

Each bombardment turn runs three rounds:

- **Rounds 1–2 (Suppression):** Attacker hits target stardock contents, then batteries. Batteries fire back each round. Armies, stored goods, and industry is shielded.
- **Round 3 (Breakthrough):** If batteries reached zero before this round, attacker hits cascade into armies, stored goods, and industry. If batteries still stand, round 3 is another suppression exchange.

Invasion uses one suppression exchange against batteries. If batteries are cleared, the softening pass targets armies only — industry and stored goods are not damaged during invasion. Orbital softening may destroy at most half of the defender's starting armies, ensuring a meaningful landing battle whenever a world began the assault with a garrison. Once armies land, ground combat in invasion and blitz proceeds in repeated simultaneous CRT rounds until one side is destroyed. The defender receives a +1 column bonus in those ground rounds, and a final exchange cannot produce a draw: the larger force entering that exchange survives with a remnant army, with exact equality favoring the defender.

13. Appendix C: Mission / Order Quick Reference

This appendix is the compact lookup version of the mission system. For full behavior explanations, examples, and tactical notes, see Section 7.

No	Abbr	Mission	Class	Trigger	Summary
00	Hd	Hold Position	Persistent	Immediate	Idle standing order; remains at current location.
01	Mv	Move Fleet	One-shot	Arrive	Travel to target sector, then stop and revert to Hold.
02	Sk	Seek Home	One-shot	Arrive	Travel to nearest owned world; retarget if it is lost en route.
03	Pa	Patrol	Persistent	Arrive	Occupy deep-space patrol sector and intercept by ROE.
04	Gs	Guard Starbase	Persistent	Arrive	Escort a starbase and remain assigned to its defense.
05	Gb	Blockade	Persistent	Arrive	Guard a world, interfere with launches and hostile access.
06	Bo	Bombard	Hostile	Next maintenance tick	Ready bombardment destroys stardock contents, defenses, armies, goods, and production.
07	In	Invade	Hostile	Next maintenance tick	Suppress batteries, soften the world, then land armies.
08	Bz	Blitz	Hostile	Next maintenance tick	Fast direct landing with higher transport and army risk.
09	Vw	View	One-shot	Arrive	Long-range scan from system edge; reports and reverts to Hold.
10	Ss	Scout Sector	One-shot	Arrive	Stealth deep-space observation; requires a Scout.
11	Sy	Scout System	One-shot	Arrive	Spy run into a world; requires a Scout.
12	Co	Colonize	One-shot	Arrive	Claim an unowned world with an ETAC; ETAC survives.
13	Jn	Join Fleet	Persistent	Merge	Chase and merge with a host fleet; abort if host is destroyed.
14	Rz	Rendezvous	Persistent	Merge	Gather fleets in one sector; lowest Fleet ID becomes host.
15	Sa	Salvage	One-shot	Arrive	Scrap ships at a world for roughly 50% of build cost.

Category key:

- **One-shot**: travel, perform an action, then revert to Hold Position.
- **Persistent**: remain armed until you replace the order or game rules invalidate it.
- **Hostile**: travel to the target world, then execute on the next maintenance tick after arrival.

14. Appendix D: Preservation and Original Sources

This edition is a playable modern Rust version of Nostrian Conquest, built with preservation discipline. The goal is to keep the original gameplay legible and playable on modern systems while documenting the exact engine rules that matter to players, operators, client authors, and future maintainers.

14.0.1. Source Hierarchy

This manual is the authoritative player manual for the Rust edition of Nostrian Conquest.

The preserved originals in `original/v1.5/` remain historical references and an ambiguity fallback:

- `ECQSTART.DOC` — Quick-start guide
- `ECPLAYER.DOC` — Detailed player manual
- `ECREADME.DOC` — Release and package information
- `ECGAME.EXE` — The original 1992 player client
- `ECMAINT.EXE` — The original yearly maintenance oracle
- `ECUTIL.EXE` — The original sysop utility for game initialization and management

14.0.2. Preservation Policy

- This manual is the authoritative player-facing manual for the Rust edition.
- The original manuals are preserved historical references and an ambiguity fallback for classic intent and terminology.
- The original DOS binaries are the acceptance oracle for compatibility work.
- The Rust engine preserves player-visible classic behavior where it matters, but it does not chase hidden byte-for-byte quirks when they do not materially affect gameplay.
- When an exact classic formula remains unrecovered, this manual documents the current engine rule explicitly rather than pretending the original value is known.
- When the preserved originals clarify an ambiguity, that clarification should be folded back into the current manuals/specs rather than left only in the legacy `.DOC` set.

14.0.3. BBS Drop File Compatibility

Modern Rust BBS hosting uses `nc-door`, not the old DOS binary path. The Rust door reads modern drop files directly, supports `DOOR32.SYS`, `DOOR.SYS`, and `CHAIN.TXT`, and avoids the brittle DOS-era parser behavior that used to make classic BBS setup so fragile. If you are a player, the practical point is simple: log into the board, launch the door, and play. If you are the sysop, see the **Sysop Manual** for host setup details.

14.0.4. This Manual

This manual combines and polishes the original documentation with the current engine reference tables in Section 11, Section 12, Section 13, and Section 15. The body text stays strategy-first; the appendices collect the exact formulas and lookup tables for readers who want the implementation-level reference.

15. Appendix E: Table Filtering and Sorting

The main list screens all use the same command-line filter system. This lets you cut a long table down to the exact rows you want instead of paging through dead weight.

One filter is active at a time on each table. If you choose a new filter, it replaces the old one. If a filter matches nothing, the table stays filtered and tells you no rows match.

15.0.1. Filtering Procedure

1. Open the table you want.
2. Press `F`.
3. Type the column code, or the shortest unique prefix, and press `Enter`.
4. Type the value for that column and press `Enter`.
5. Type `all` at the column prompt to clear the current filter.

NOTE: `Q` or `Esc` cancels the current filter prompt without changing the active filter.

If your prefix matches more than one code, the command line stays open and shows the matching codes so you can narrow the entry.

15.0.2. Value Rules

- **Text columns:** Enter plain text. Matching is case-insensitive and looks for that text anywhere in the cell.
- **Number columns:** Enter a bare number for an exact match, or use `>`, `>=`, `<`, `<=`, `=`, or `!=`.
- **Coordinate columns:** Enter `xx,yy` for one exact sector, or `xx,yy/r` for a radius filter.
- **Database unknown values:** Enter `?` to match worlds where that value is still unknown.

Examples:

- Fleet list, `ord`: `holding`
- Fleet list, `sel`: `yes`
- Fleet list, `roe`: `>=4`
- Planet list, `coo`: `12,7/3`
- Total planet database, `own`: `#3`
- Total planet database, `max`: `>=100`

15.0.3. Sorting Procedure

1. Open the table you want.
2. Press `S`.
3. Press the sort key shown on the command line.
4. Press the same sort key again to flip `ASC`/`DESC`.

The active sort and active filter both appear in the table title.

15.0.4. Fleet List Codes

Code	Column	Notes
`id`	Fleet ID	Fleet number
`loc`	Location	Current sector
`ord`	Order	Also accepts `holding`, `moving`, and `combat`
`tar`	Target	Mission target sector
`spd`	Speed	Current speed
`eta`	ETA	Text match on the ETA column
`roe`	ROE	Rules of engagement value
`ars`	Armies	Loaded armies
`shi`	Ships	Ship and force summary text
`sel`	Selected	Use `yes` or `no`

15.0.5. Planet List Codes

Code	Column	Notes
`coo`	Coord	Planet coordinates
`pla`	Planet	Planet name
`max`	Max	Maximum production
`cur`	Curr	Current production
`trs`	Points	Stored treasury points
`bdg`	Bdgt	Build budget
`rev`	Rev	Revenue
`gro`	Grow	Growth
`bui`	Queue	Build queue size
`sta`	Dock	Docked ships
`sbs`	SBs	Friendly starbases
`ars`	ARs	Planet armies
`gbs`	GBs	Ground batteries

15.0.6. Total Planet Database Codes

Code	Column	Notes
`coo`	Coord	World coordinates
`pla`	Planet	Known planet name
`own`	Owner	Known owner text such as `#3` or `Unowned`
`max`	Max	Known maximum production
`see`	Seen	Last year seen
`ars`	ARs	Known armies
`gbs`	GBs	Known ground batteries
`sbs`	SBs	Known starbase count
`cur`	Curr	Known current production
`trs`	Points	Known stored points
`sco`	Scout	Last scout year